

2.0 Project Overview

Application Name: Health Habit Tracker

Purpose: A GUI-based health tracking system designed to help users monitor daily health habits while contributing to Sustainable Development Goal 3 (Good Health and Well-being).

Target Audience: Individuals seeking to improve physical and mental health through consistent tracking of water intake, sleep, exercise, meditation, and reading.

Alignment with SDG 3: This application directly supports "Good Health and Well-being" by enabling users to track and improve their daily health habits, including water intake, sleep, exercise, meditation, reading, and supplement consumption.

In terms of software design, the system demonstrates object-oriented complexity through abstraction, inheritance, polymorphism, the factory design pattern, database persistence, and event-driven GUI design using ScalaFX. These design decisions ensure the application is modular, extensible, and maintainable while meeting the learning outcomes of object-oriented programming.

Technology Stack: Scala 3, ScalaFX 21, Java 21, and Apache Derby Database.

3.0 UML Diagram

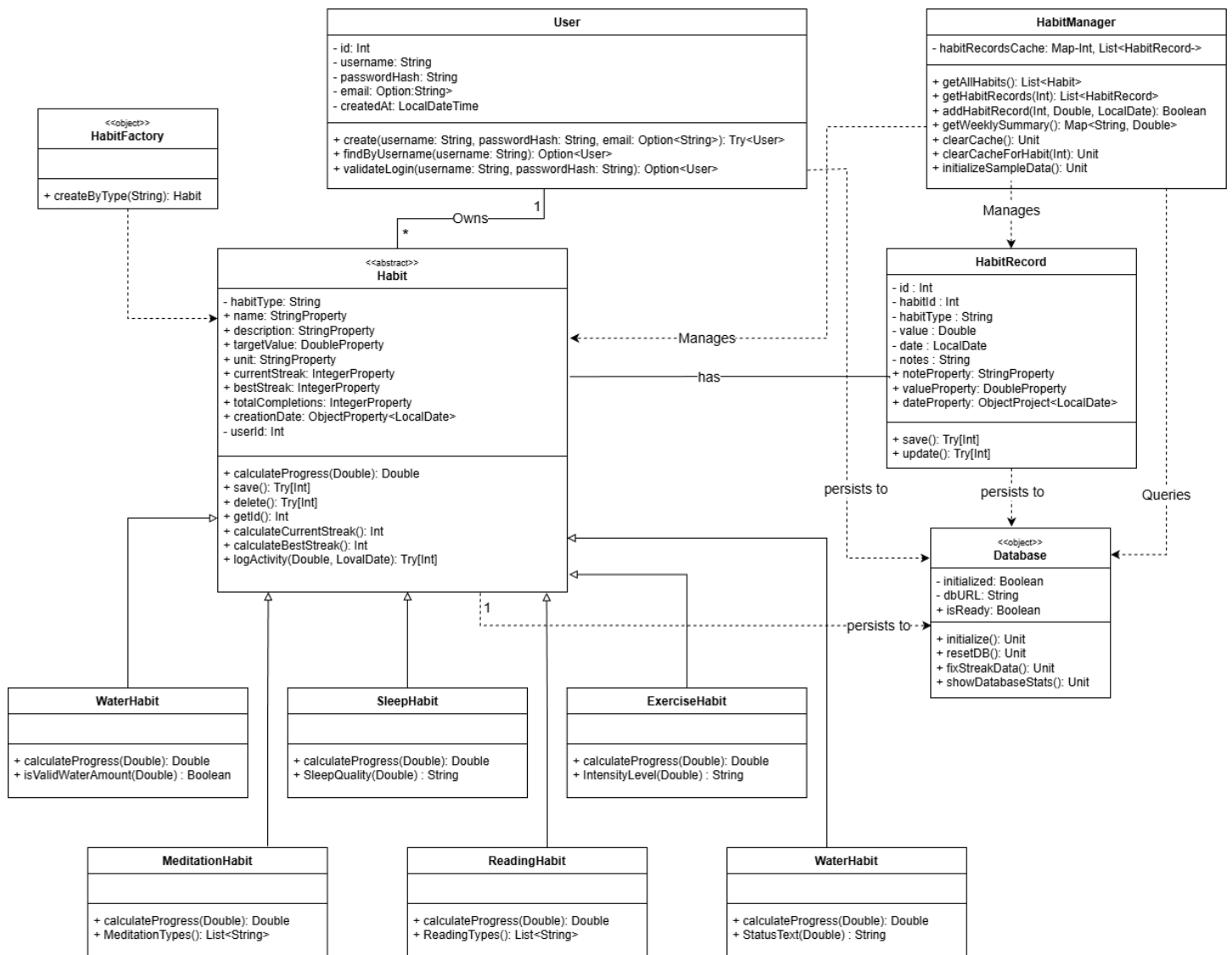


Figure 3.1: UML Class Diagram for Health Habit Tracker

At the center of the system is the Habit abstract base class, which acts as a template for every habit in the app. It stores shared information like the name, description, and target goal using ScalaFX properties. Because it is an abstract class, it defines a required method called calculateProgress but leaves the specific math for that calculation to the individual habit types.

To handle different health activities, I used Inheritance to create six specific subclasses: WaterHabit, SleepHabit, ExerciseHabit, MeditationHabit, ReadingHabit, and SupplementHabit. Each inherits from the main Habit class but adds its own unique logic for progress and specific helper methods. To keep data private, I

included a User class with a "one-to-many" association, ensuring each user only interacts with their own personal data.

For the technical implementation, I applied the Factory Pattern through the HabitFactory class to create the right habit objects based on text input. I also created a HabitManager to organize these objects and use a "cache" for fast data loading. Individual logs are stored as a HabitRecord, which has a "many-to-one" relationship with a habit. All data is saved permanently using a Database singleton. Finally, the user interface is managed by controllers that use Polymorphism to display and log information for any habit type using shared screens.

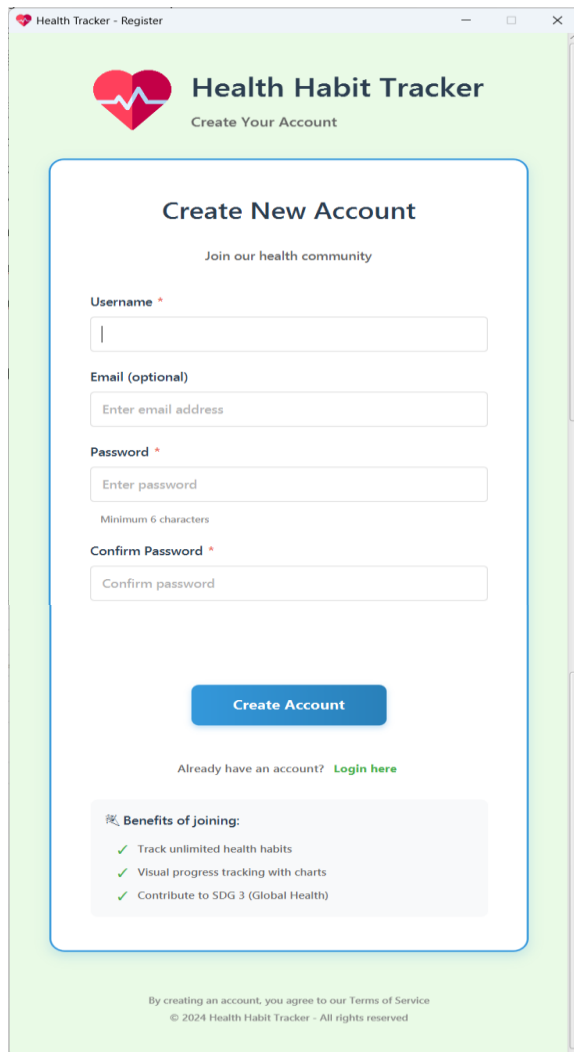
4.0 Feature Description

The Health Habit Tracker program is confirmed to work for the following 4 features:

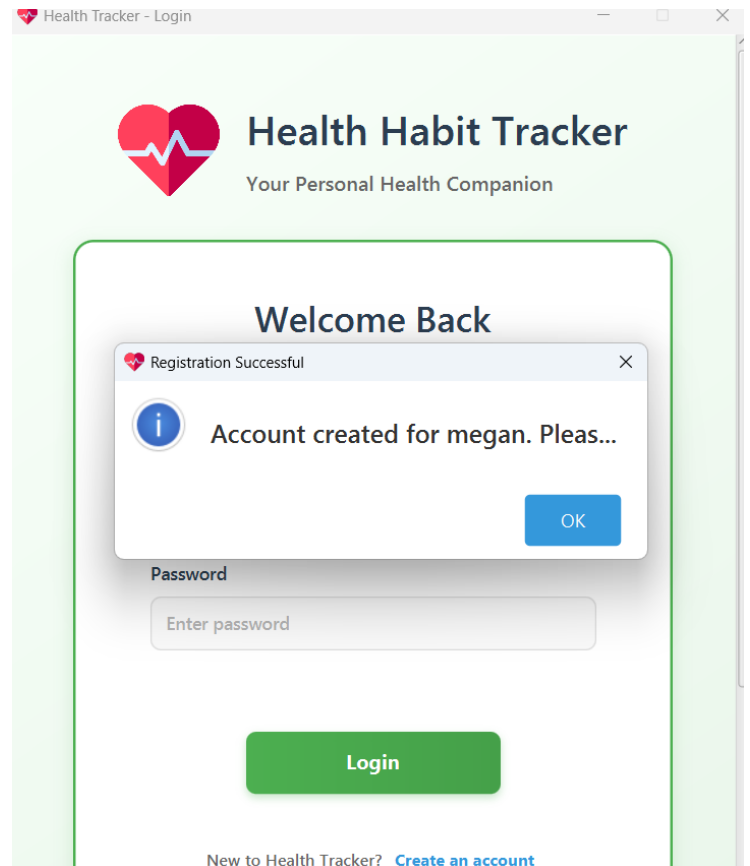
1. User Authentication (Security): Users can register new accounts and log in securely. The system uses SHA-256 hashing to protect passwords via the PasswordHasher utility.
2. Habit Management: Users can create, edit, and delete habits. Each habit is saved in the Derby database and linked to the specific user_id of the logged-in user.
3. Daily Activity Logging: Through the HabitLogDialog, users can input daily values (e.g., 2000ml of water). This data is stored as a HabitRecord and immediately reflects in the progress calculations. Basic input validation is applied to ensure that users cannot enter negative or invalid values when logging daily activities.
4. Visual Analytics: The system provides two distinct charts:
 - LineChart: Displays a 7-day progress trend for the selected habit.
 - BarChart: Compares the weekly average progress of all habits against their set targets.

4.1 App Screenshot and explanation

4.1.1 Register



The screenshot shows the 'Health Tracker - Register' page. At the top, there is a red heart logo with a white pulse line and the text 'Health Habit Tracker' and 'Create Your Account'. Below this is a white box with a blue border containing the 'Create New Account' form. The form has the sub-header 'Join our health community' and four input fields: 'Username *', 'Email (optional)' (with placeholder 'Enter email address'), 'Password *' (with placeholder 'Enter password' and a note 'Minimum 6 characters'), and 'Confirm Password *' (with placeholder 'Confirm password'). A blue 'Create Account' button is at the bottom of the form. Below the button, it says 'Already have an account? [Login here](#)'. At the bottom of the white box, there is a section titled 'Benefits of joining:' with three green checkmarks: 'Track unlimited health habits', 'Visual progress tracking with charts', and 'Contribute to SDG 3 (Global Health)'. At the very bottom of the page, small text reads: 'By creating an account, you agree to our Terms of Service' and '© 2024 Health Habit Tracker - All rights reserved'.

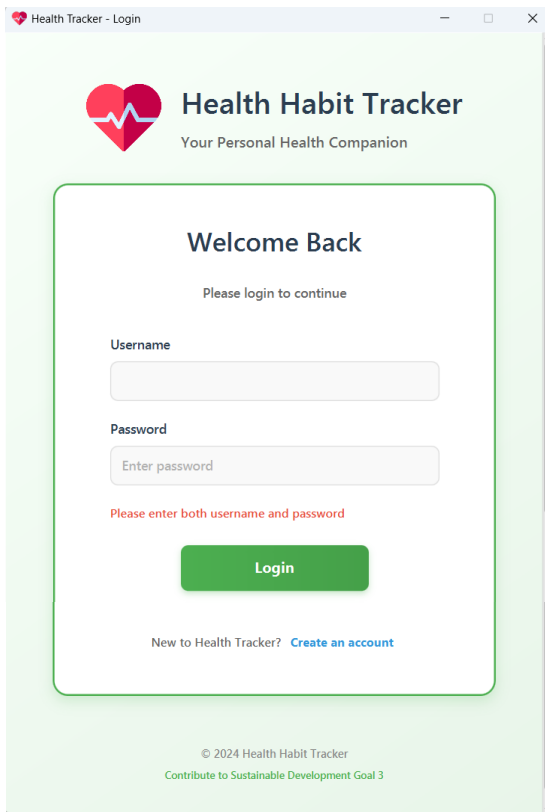


The screenshot shows the 'Health Tracker - Login' page. At the top, there is a red heart logo with a white pulse line and the text 'Health Habit Tracker' and 'Your Personal Health Companion'. Below this is a white box with a green border containing the 'Welcome Back' login form. The form has a 'Password' label and an input field with placeholder 'Enter password'. A green 'Login' button is at the bottom of the form. Below the button, it says 'New to Health Tracker? [Create an account](#)'. Overlaid on top of the login form is a white dialog box with a blue border. The dialog box has a title bar that says 'Registration Successful' with a close button (X). Inside the dialog, there is a blue information icon (i) and the text 'Account created for megan. Pleas...'. A blue 'OK' button is at the bottom right of the dialog.

Figure 4.1 & 4.2 Registration page

This is the Register page where new user can create their account. They need to fill in their username, email and password. The email is optional, meaning user can choose not to fill in their email. After done filling their username and password, user press “Create Account” button to create account. A Registration successful dialog box will appear to notify the user that they have created their account successfully.

4.1.2 Login



This is the Login page. After the user creates their account, it will automatically jump to the Login page for the users to Login. User have to key in their username and password they created and click on “Login” button. If the user haven’t have an account yet, they can click on the “Create an account” button at the bottom of the Login page to create account. Users must key in both Username and password. If they did not, there will have a read error message – “Please enter both username and password.”

Figure 4.3 Login Page

4.1.3 Menu bar

4.1.3.1 Navigation

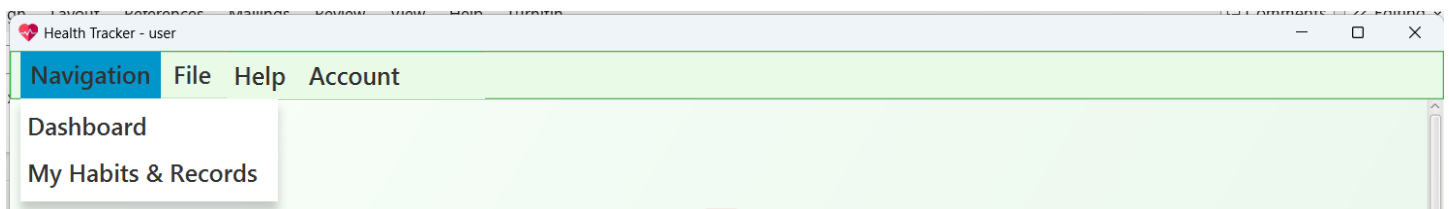


Figure 4.4 Navigation Menu

This is the navigation bar. Users can go back to dashboard or see their habit records no matter which page they are at. For example, if the users are at their habit record page and wants to go back to the dashboard, they can press this “Navigation” button and click on “Dashboard”.

4.1.3.2 File

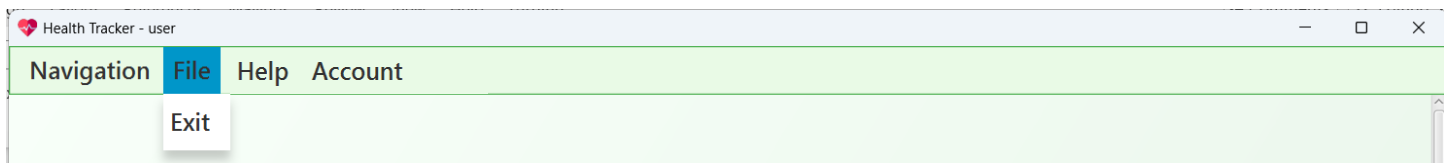


Figure 4.5 File Menu

This is the File Menu; users can press on this “Exit” button or click on the “X” on the top right to close the whole program.

4.1.3.3 Help

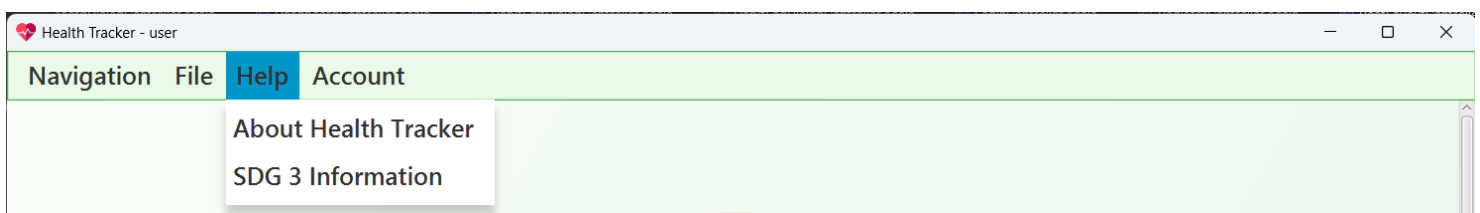


Figure 4.6 Help Menu

This is the Help Menu; users can see information about this app – Health Habit Tracker, and the SDG related to it – SDG 3 Good Health and Well Being.

4.1.3.4 Account

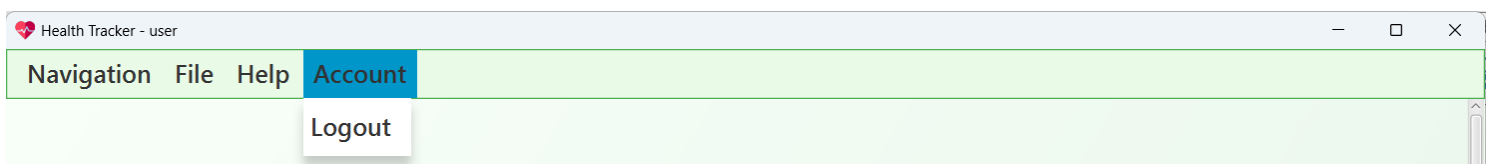


Figure 4.7 Account Menu

This is the Account Menu; users can press this to log out from their account. After pressing this logout button, it will direct user back to the login page.

4.1.4 Dashboard

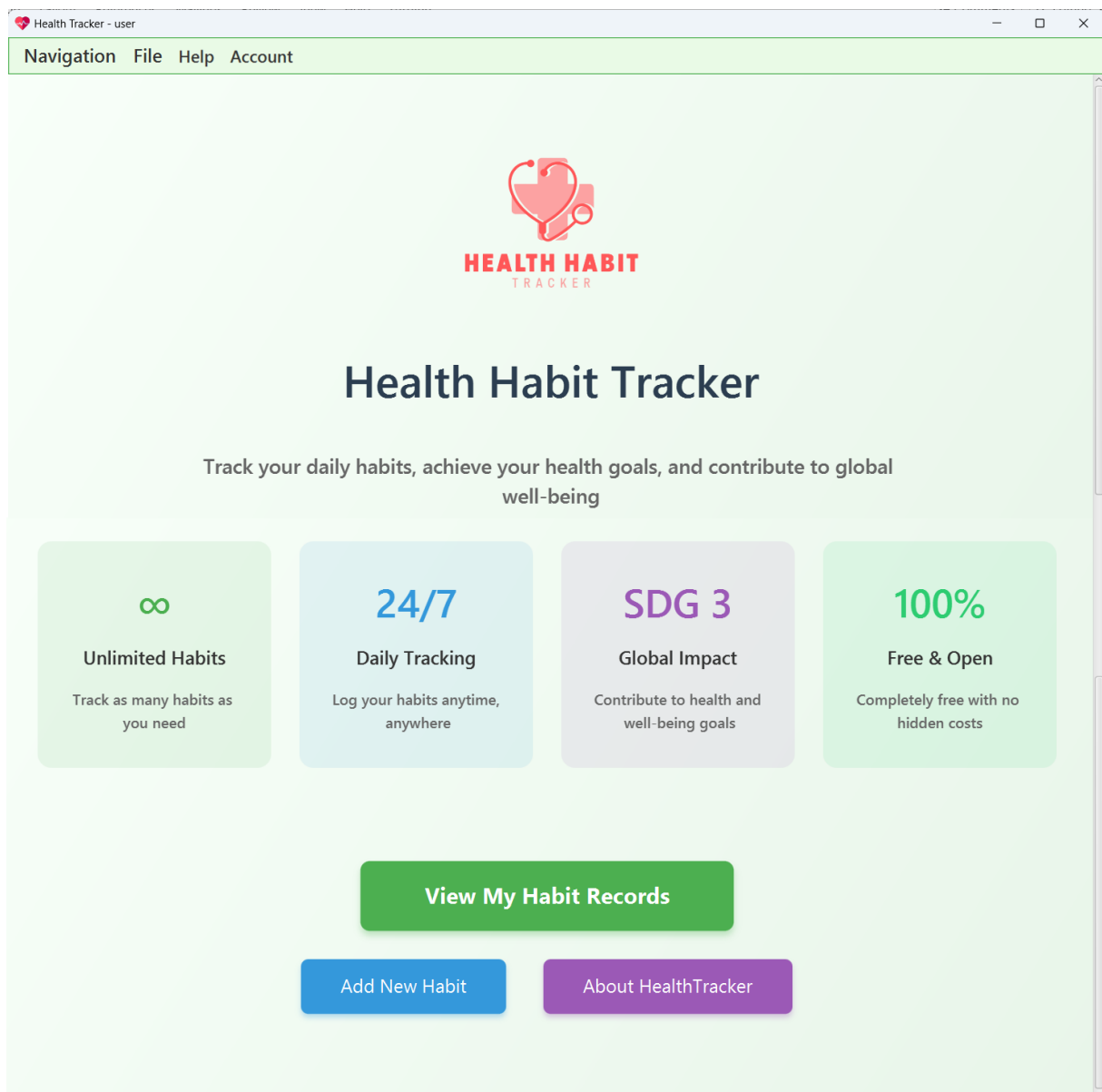


Figure 4.8 Dashboard

This is the dashboard. In the dashboard, it briefly explains what this app do. Below of the dashboard page, there are 3 buttons. The first button is “View My Habit Record”. Users can enter the Habit Record page from here as well.

The second button is “Add new Habit” button. Other than going in to the Habit Record page to add new habits, users can also add their new habit here.

The third button is the “About HealthTracker” button. Instead of going to the about page from the menu bar, users can also see the information about this app by pressing this button.

4.1.5 Habit Record (AddRecordDialog)

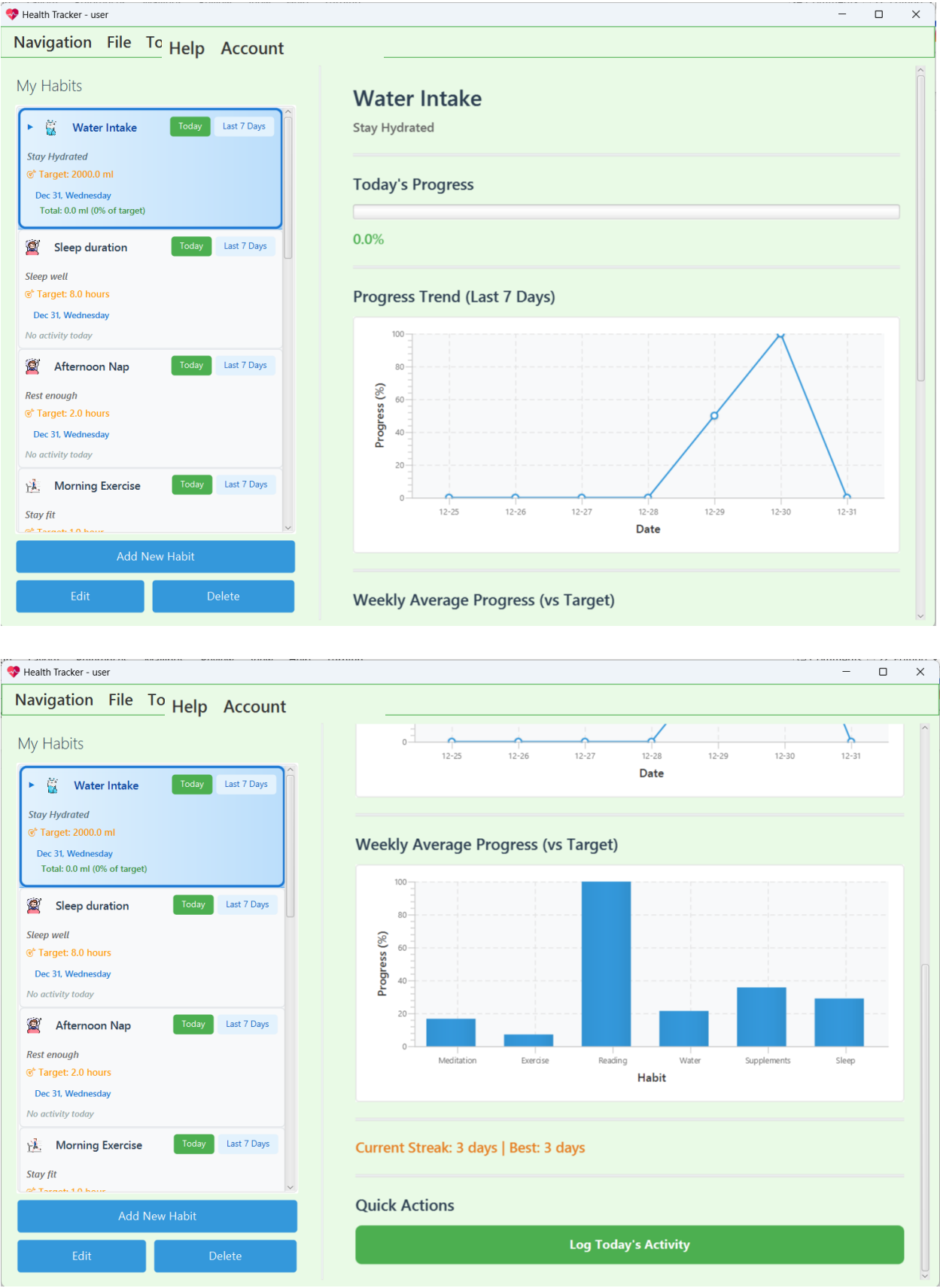


Figure 4.9 and 4.10 Habit Record Page

This is the Habit Record Page. All the habits created and logged by the users are recorded here. Users can see their previous and current records at the left side of the page. To see the previous records, users just need to press on the “Last 7 Days” button on each habit. If there are no previous habit records, the “Last 7 Days” button will be disabled.

On the right side of the page shows the selected habit’s name and description. Below is the progress bar for the selected habit, it takes note how much progress have made from the target for each habit. This is how the progress bar percentage is calculated: $(\text{the amount done} / \text{the target amount}) * 100\%$. If the target for the habit “Water” is 1000ml, but the user didn’t log any activity, the progress bar remains 0. If the user logged 500ml, the progress bar will become 50%.

Below the progress bar is the progress trend, it shows daily progress percentages for a selected habit over the last 7 days. For each day, it sums all logged values, divides by the habit's daily target, and caps the result at 100%. Days with no entries show as 0%. This creates a line showing day-to-day consistency.

Other than that, there is also a weekly summary chart where users can see their weekly average progress. It calculates average daily performance for each habit type (water, exercise, sleep, reading, meditation, supplement). For each type, it averages daily totals across all habits of that type over the past week, then converts this average to a percentage of the daily target. Each bar shows how well that category was maintained overall during the week.

There are also streak calculations for the habits. The streak calculation tracks consecutive days a user meets or exceeds their habit target. Each day, the system checks if the daily total equals or surpasses the target value. If yes, the current streak increments; if no, it resets to zero. The best streak stores the highest consecutive value ever achieved. Streaks are updated in real-time whenever new activity is logged, providing motivational feedback on consistency over time.

Lastly, the users can log their activity by pressing on the “Log Today’s Activity” button. This will record the amount done of each habit and will display on the habit list under each habit. If the user log 3 times of that habit, it will appear as a list under that day of each habit.

4.1.6 Add New Habit (HabitEditDialog)

The screenshot shows the 'Add New Habit' dialog box overlaid on the 'My Habits' page of the 'Health Tracker' application. The dialog box has a title bar 'Add New Habit' and contains the following fields:

- Habit Type:** A dropdown menu with 'Water' selected.
- Name:** An empty text input field.
- Description:** An empty text input field.
- Target:** A text input field containing '0.0'.
- Unit:** An empty text input field.

At the bottom of the dialog box are two buttons: 'Save' and 'Cancel'.

The background application window shows a navigation bar with 'Navigation', 'File', 'Help', and 'Account'. The 'My Habits' section lists several habits:

- Water Intake:** 'Stay Hydrated', Target: 2000.0 ml, Dec 31, Wednesday, Total: 0.0 ml (0% of target).
- Sleep duration:** 'Sleep well', Target: 8.0 hours, Dec 31, Wednesday, No activity today.
- Afternoon Nap:** 'Rest enough', Target: 2.0 hours, Dec 31, Wednesday, No activity today.
- Morning Exercise:** 'Stay fit', Target: 1.0 hour.

At the bottom of the application window, there is a 'Quick Actions' section with a 'Log Today's Activity' button.

Figure 4.11 Add New Habit

This is the form that allows users to key in their habit and the target for that habit. Users click on the “Add New Habit” button and this form will pop out. After keying everything inside, the users press “Save” to save their new habit and the habit will be displayed in the habit list.

4.1.7 Log Activity (HabitLogDialog)

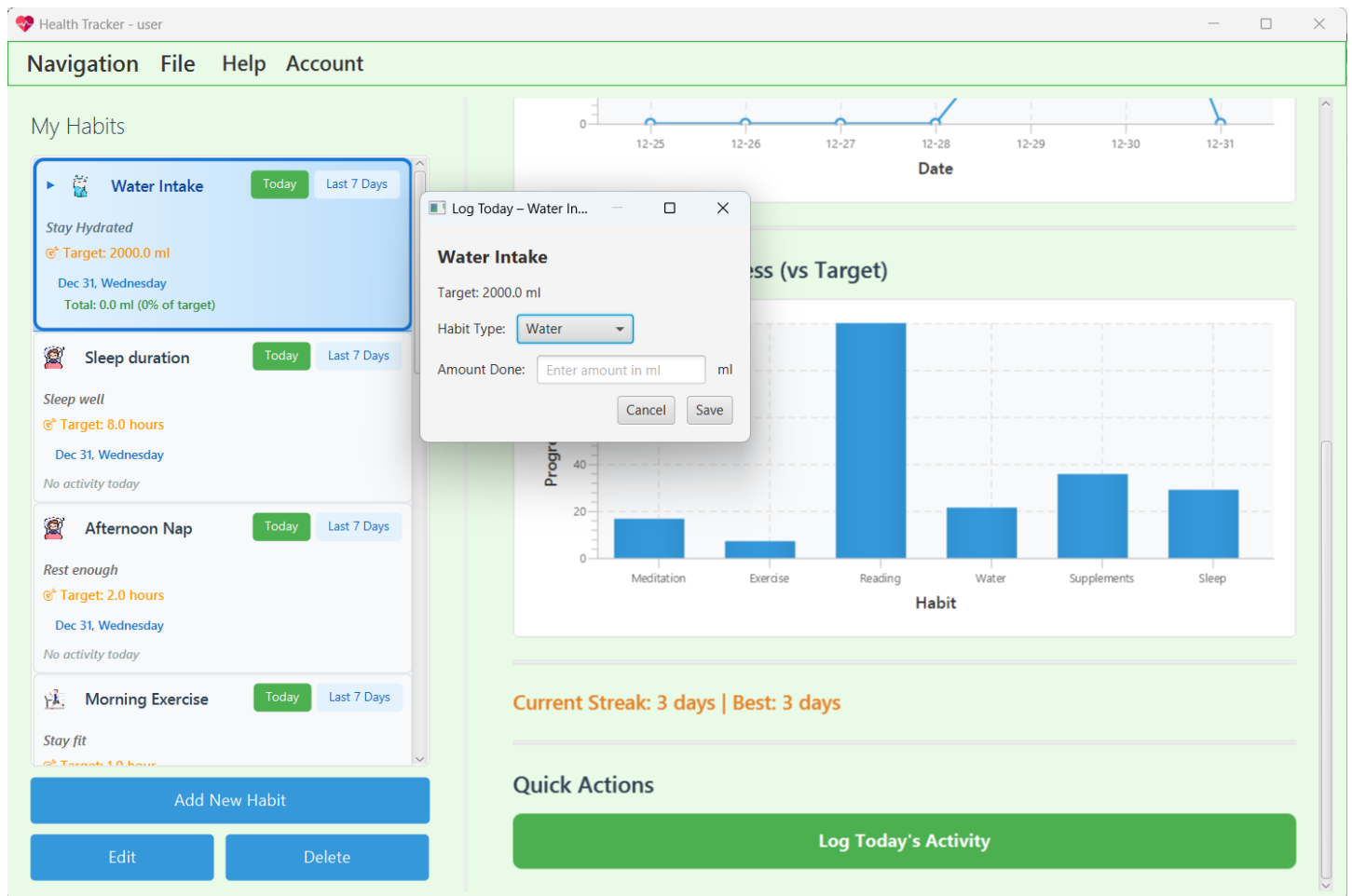


Figure 4.12 Log Activity

This is the form that allows users to log their activity. Users click on the “Log Today’s Activity” button to record the amount done of that habit. This allows the users to keep track of their progress for each habit. Logged activities cannot be edited but have no limits on logging. If the user accidentally logged less amount, they could log again. If the accidentally logged more amount, they can minus off from the next log or increase the target by editing it.

4.1.8 Edit Habit

Health Tracker - user

Navigation File Help Account

My Habits

- Water Intake** Today
Stay Hydrated
Target: 2000.0 ml
Dec 31, Wednesday
Total: 0.0 ml (0% of target)
- Sleep duration** Today
Sleep well
Target: 8.0 hours
Dec 31, Wednesday
No activity today
- Afternoon Nap** Today
Rest enough
Target: 2.0 hours
Dec 31, Wednesday
No activity today
- Morning Exercise** Today Last 7 Days
Stay fit
Target: 1.0 hours

Add New Habit Edit Delete

Edit Habit

Habit Type: Water

Name: Water Intake

Description: Stay Hydrated

Target: 2000.0

Unit: ml

Save Cancel

Date

12-27 12-28 12-29 12-30 12-31

Habit

Reading Water Supplements Sleep

Current Streak: 3 days | Best: 3 days

Quick Actions

Log Today's Activity

Figure 4.13 Edit Habit Record

This is the form to allow the users to edit their habit name, description, target of target unit. Users must select on the habit they want to edit, click on the “Edit” button, change the information and click “Save”. The updated habit record will overwrite the previous record and display the updated habit record in the habit list.

4.1.9 Delete Habit

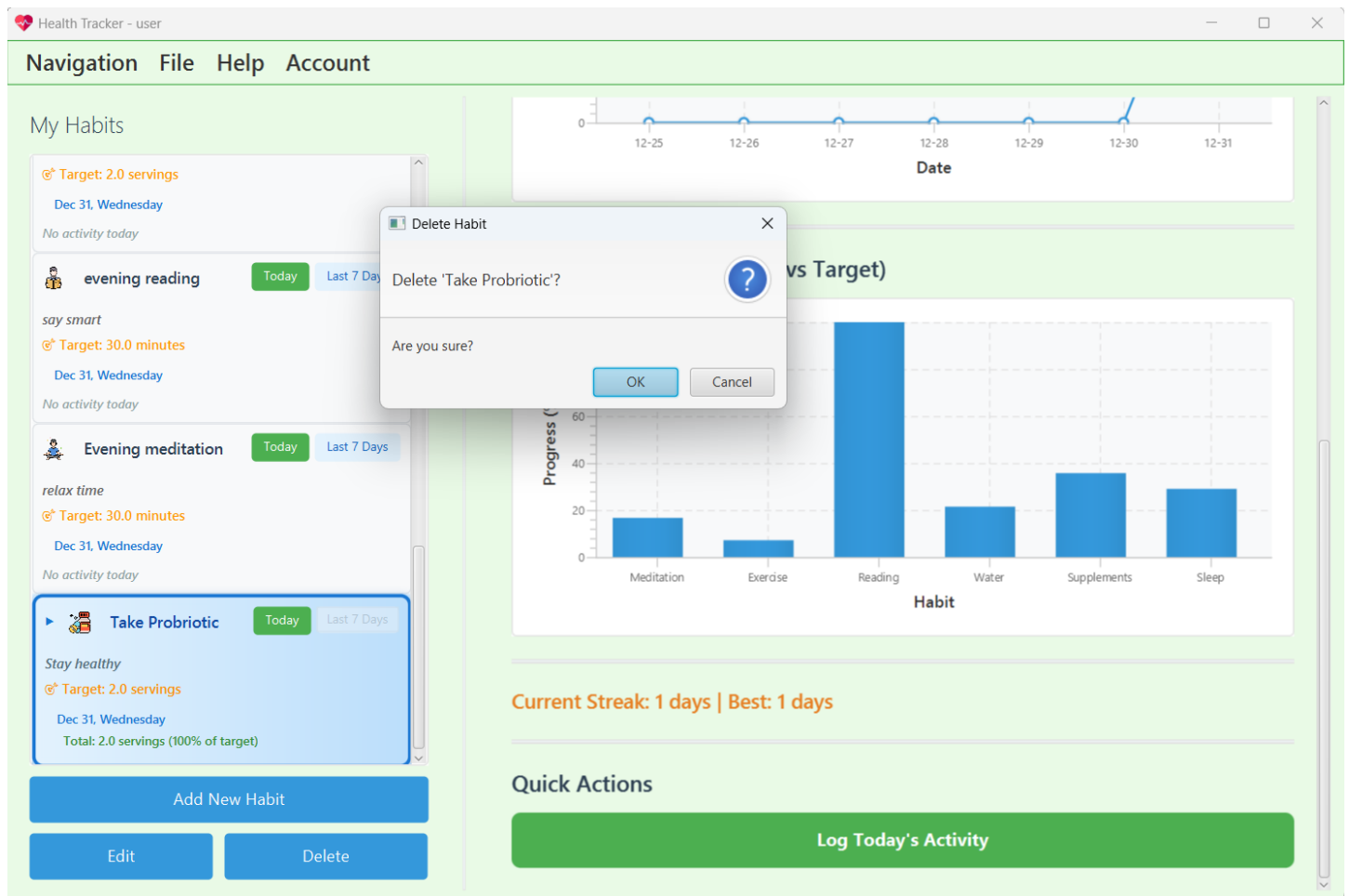


Figure 4.14 Delete Habit Record

The users can delete their habit by just clicking on the “Delete” button. A pop-up window will appear to make confirmation with the user in case the user accidentally clicked it. After the user click on “Ok” button, that selected habit record will be deleted permanently.

4.1.10 About Health Habit Tracker

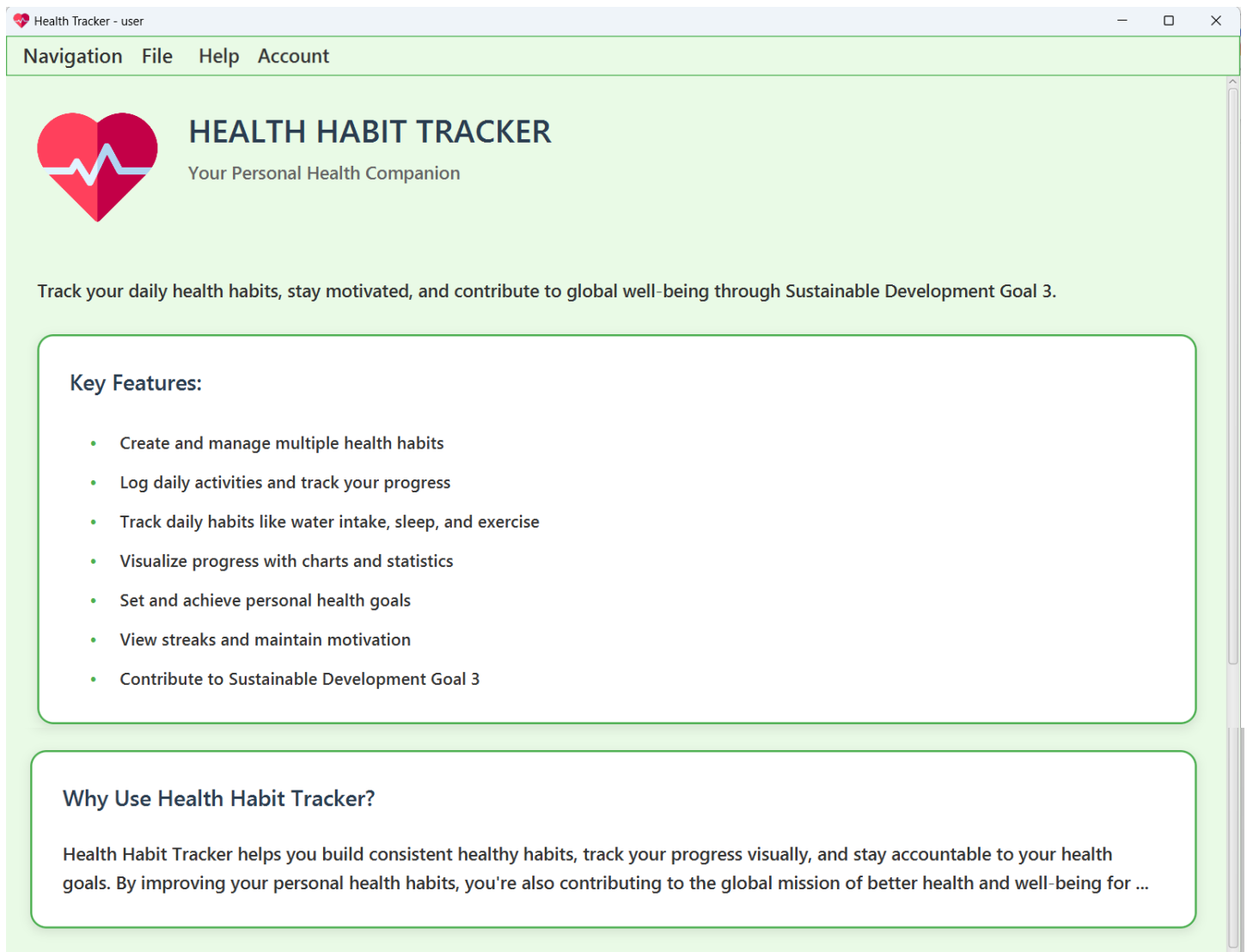


Figure 4.15 About Health Habit Tracker Page

This is the page about this app. It explains the features of the app and why users should use this app. This is a small brief for users to refer to.

4.1.11 About SDG 3 Information

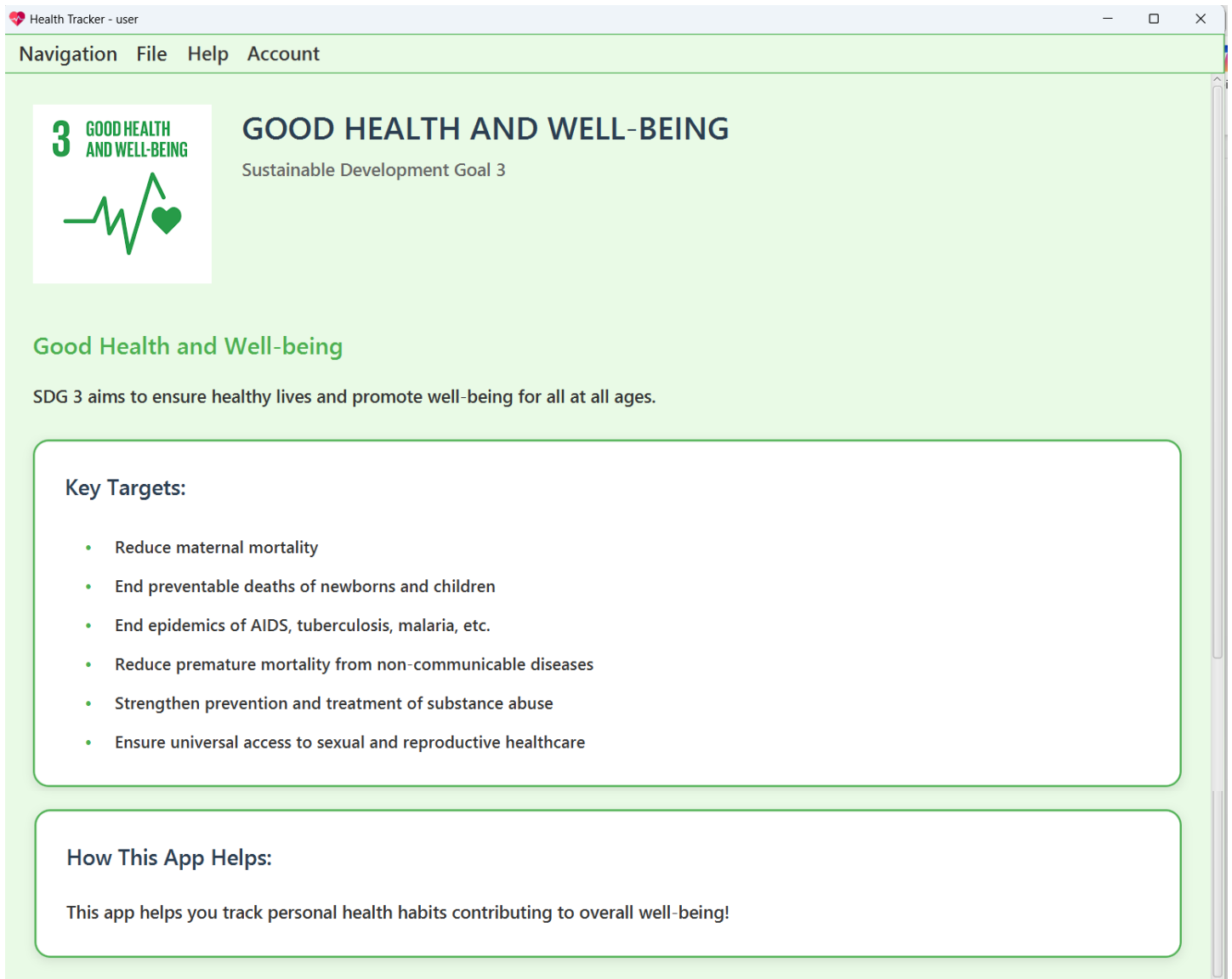


Figure 4.16 SDG 3 Information Page

This is the page about the SDG related to this app – SDG 3. It explains the key targets of SDG 3 and how this app can help to meet the target.

4.1.12 Logout

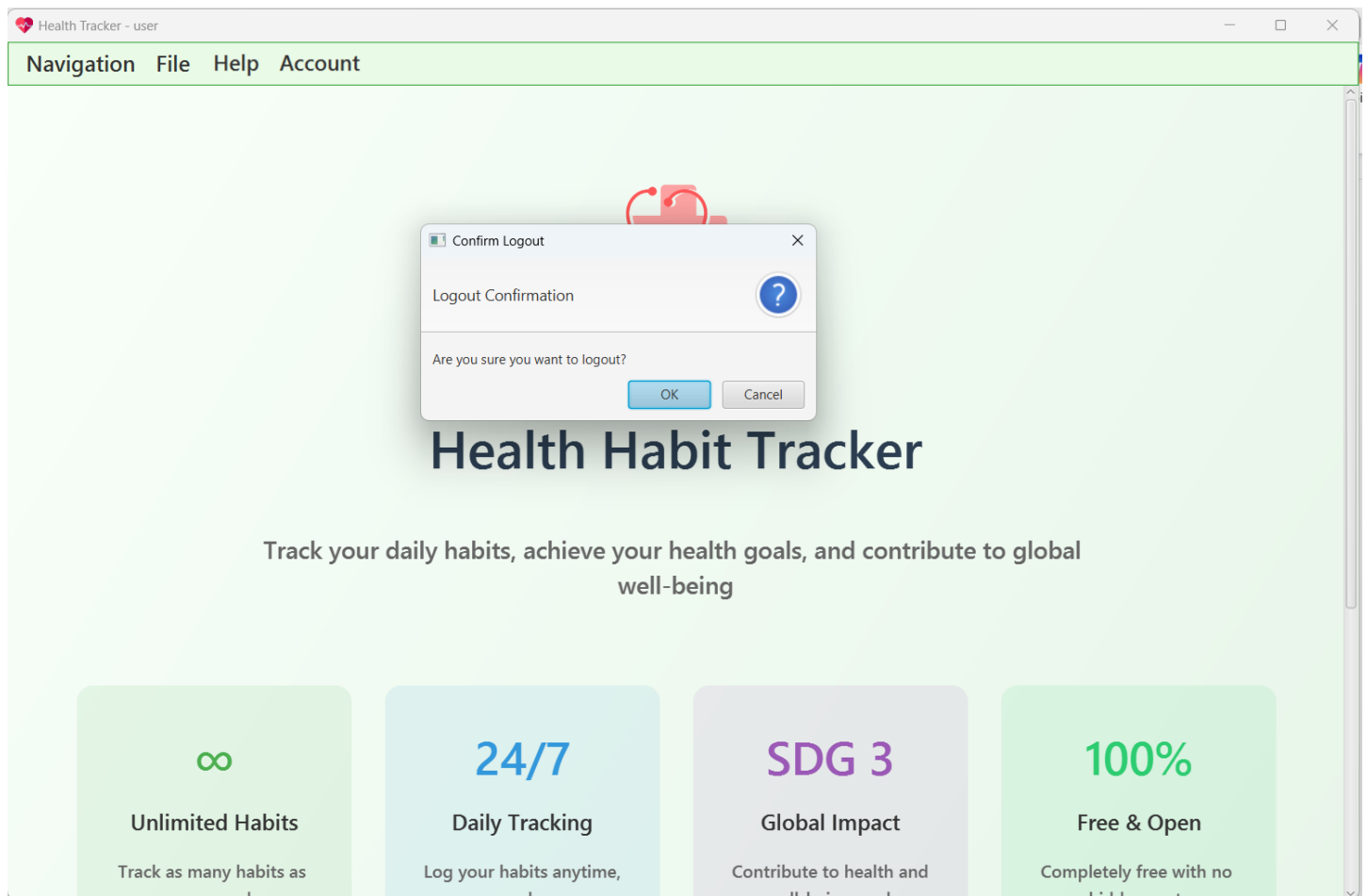


Figure 4.17 Logout Page

When users want to logout from their account, they can press the “Account” button on the menu bar, no matter which page they were at, and press “Logout”. A pop-up confirmation box will appear to make confirmation with the user. After pressing ok, the user will be logged out and will go back to the login page.

5.0 Personal Reflection

During the development of the Health Habit Tracker, I applied several core Object-Oriented Programming (OOP) concepts that were taught in class. First, I used **Abstraction** by creating an abstract base class called `Habit`. This allowed me to define a common structure for all habits while hiding complex implementation details from the rest of the system. I then applied **Inheritance** by creating specific habit types such as `WaterHabit` and `SleepHabit`, which reused shared behaviour from the base class while adding their own specific logic.

In addition, **Polymorphism** was implemented in the user interface, particularly in the `ListView`, where different habit types can be handled as general `Habit` objects. This made the system more flexible and easier to manage. I also applied **Encapsulation** by using `ScalaFX` property types to protect data fields and ensure that the user interface and database interact with data through controlled methods. **Generic programming** was used when handling collections such as `List[Habit]` and `ObservableBuffer[Habit]`, allowing type-safe and flexible management of different habit objects.

`ScalaFX` was chosen as the GUI framework because it provides a type-safe `Scala` wrapper over `JavaFX`. This allows reactive UI binding through properties and clean event handling, while remaining compatible with `JavaFX 21`. As a result, it was easier to build a responsive and maintainable graphical user interface for the application.

Throughout the project, I encountered several technical challenges that required problem-solving. One major issue was managing the database connection so that new user accounts could be created without removing existing habit data from previous sessions. I solved this by implementing a safe database setup method that checks whether tables already exist before creating them. Another challenge was that the progress charts did not always refresh immediately after logging a new activity. To address this, I created a `forceCompleteRefresh` function in the controller to clear cached data and reload the updated information on the main screen.

In evaluating my work, I believe the main strength of this application is its **modular design**. By using the Factory Pattern to create habit objects, new habit types can be added easily without changing the core system logic. The user interface is also a strength, as custom CSS and visual charts were used to improve usability and

give the application a more professional appearance. However, a limitation of the system is that habits currently use fixed units, such as minutes or millilitres. Given more time, I would improve the system by allowing users to customise measurement units, making the application more flexible for different users.